Adilet Shynggys

Astana, Kazakhstan | adilet.shynggys@nu.edu.kz | +7 700 959 0833 | adiletshynggys.github.io/portfolio |
linkedin.com/in/adilet-shynggys

Research Interests

Nuclear reactor thermal-hydraulics, subcooled boiling, and critical heat flux; interfacial and multiphase fluid mechanics; capillary and contact-line dynamics; reduced-order modeling of two-phase transport in confined geometries.

Education

Nazarbayev University — Astana, Kazakhstan
B.Eng. in Mechanical and Aerospace Engineering — *Expected June 2027*

- **GPA: 3.86 / 4.00** — ranked **1st** in cohort; Dean's List, four consecutive semesters.
- Relevant coursework: Fluid Mechanics; Engineering Thermodynamics; Numerical Methods; Differential Equations and Linear Algebra; Engineering Dynamics I–II; Structural Mechanics; Materials and Manufacturing; CAD. In progress: Heat and Mass Transfer; Mechanical Systems Design; Aerospace Propulsion.

The University of Hong Kong — Hong Kong SAR
Visiting Exchange Student, Department of Mechanical Engineering — *Spring 2026*

- Coursework in Energy Conversion Systems, Design and Manufacture, and Additive Manufacturing; completed two design projects during the exchange (see Engineering Design Projects) and collaborated with Prof. Wei Yan on curriculum design for an undergraduate course.

Research Experience

Undergraduate Researcher — Thermal-Hydraulics and Multiphase Flow — June 2026 – Present
Department of Mechanical and Aerospace Engineering, Nazarbayev University • Advisor: Prof. Amgad Salama

- Extending a semi-analytical framework for wetting and non-wetting liquid penetration through cylindrical and tapered capillaries, resolving meniscus velocity, dynamic contact angle, and film thickness.
- Formulating and numerically integrating the coupled momentum–capillarity equations governing contact-line motion in MATLAB and Python; benchmarking against the Lucas–Washburn and Bosanquet limits.
- Developing reduced-order models of capillary-driven two-phase transport that capture the dominant interfacial physics at far lower cost than 3D simulation.
- Designing tapered SU-8 microchannels for photolithographic fabrication on silicon to test the model experimentally: the width gradient creates a Laplace-pressure asymmetry driving wetting liquids toward the narrow end and non-wetting liquids toward the wide end.
- Connecting this geometry-driven transport to passive liquid return in two-phase thermal management and to rewetting in heated channels.

Undergraduate Researcher — Triboelectric Energy Conversion — Jan. – Dec. 2025
Energy Conversion Laboratory, Nazarbayev University

- Fabricated triboelectric nanogenerators for self-powered wearable sensing (e.g., robotic-hand and haptic feedback systems, hiking/outdoor gear) via electrospinning and silicone casting; varied surface morphology and material pairing to raise output power density by **15–20%**.
- Built the electrical characterization workflow (open-circuit voltage, short-circuit current, load sweeps) and processed the resulting datasets in OriginPro and MATLAB.
- Contributed experimental data and analysis to a co-authored article in *Small* and to an award-winning conference poster.

Publications and Conference Presentations

[1] S. Faramarzi, S. Yavari, U. Ospanova, **A. Shynggys**, et al. "Polarity-Engineered, Tribo-Positive Imidazolium-Mediated Crosslinked 6FDA-Durene Films for Triboelectric Nanogenerator," *Small*, 2026, art. e74486. `https://doi.org/10.1002/smll.74486`

[2] **A. Shynggys**, S. Yavari, G. Kalimuldina. "Green-Synthesized Ag Nanoparticles Incorporated into High-Performance Triboelectric Wearable Sensors." Presented as a poster at the Belt and Road Initiative Conference, Astana, 2025 (**Best Research Poster Award**), and as an abstract at the 13th International Conference on Nanomaterials and Advanced Energy Storage Systems (INESS-2025), Nazarbayev University, Astana, Aug. 2025.

Engineering Design Projects

Nuclear Power Plant Conceptual Design 2026 – 2027

Senior Capstone Project, Nazarbayev University • Team of 4 • In progress

- Leading a four-student team on the conceptual design of a **VVER**-type pressurized-water reactor; owning core thermal-hydraulic analysis — power rating, heat balance, primary/secondary heat removal.
- Identifying DNB as the limiting failure mode under subcooled boiling; planning a critical heat flux (CHF) margin analysis via power shaping, mass flux, and subcooling.
- Selecting structural, fuel, and coolant-boundary materials against thermal, mechanical, and irradiation requirements.

Ergonomic Train-Driver Seat Design 2025

*Mechaton Engineering Case Competition, sponsored by Wabtec Corporation • Team of 3 • **1st Place***

- Designed and structurally analyzed a driver's seat for 14-hour locomotive shifts, certifying it against **GOST 33330-2015** under static/dynamic fatigue loading; built a low-cost demonstration from simple materials to present results to judges.

Design Projects — The University of Hong Kong Spring 2026

Prof. Wei Yan (transmission design) • Prof. Homayoon Kazerooni, UC Berkeley (art installation)

- *Multi-Stage Transmission (Gearbox) Design*: Sized shafts using the **ASME** combined bending–torsion criterion and selected gear modules, keyways, and bearings to meet the project's specified safety margin and fatigue life.
- *Interactive Public Art Installation*: Modeled flower-petal forms in SolidWorks to specified dimensional requirements, arranging multiple petals to avoid geometric collisions and verifying structural stability, then 3D-printed each piece on a Bambu Lab A1.

Industry Experience

Product Support Engineering Intern June – July 2025

Wabtec Corporation — Kazakhstan

- Analyzed thermal performance of locomotive traction motors and supported root-cause review of field temperature excursions.
- Audited shop-floor manufacturing against engineering drawings with a fellow intern, reconciling GOST (shop) with ASME (Wabtec's U.S. standard); traced an insulation-usage inconsistency to climate differences between export markets (e.g., uninsulated units for India vs. insulated units for Kazakhstan's winters).

Awards and Honors

- **Presidential Scholarship of the Republic of Kazakhstan** — national merit scholarship 2026–2027
- **NU President's Scholarship** — full-tuition university merit award 2023–2027
- **Yessenov Foundation Scholarship** — top 20 of 500+ applicants 2026–2027
- **Best Research Poster**, Belt and Road Initiative Conference 2025
- **1st Place**, Mechaton Engineering Case Competition (sponsored by Wabtec) 2025
- **Champion**, NU AIChE Engineering Challenges 2025

Technical Skills

Programming	MATLAB, Python, Mathematica, LaTeX
Modeling & Numerics	Reduced-order modeling of multiphase transport; nonlinear ODE integration; validation against analytical limits
Simulation & CAD	SolidWorks (CSWA certified), OriginPro
Experimental	Electrospinning, silicone casting, 3D printing, electrical device characterization
Design Codes	ASME shaft design, GOST 33330-2015
Languages	Kazakh (native), Russian (fluent), English (IELTS Overall Band 8.0)

Teaching, Leadership, and Service

- **Physics Tutor**, Nazarbayev University — 20+ first-year students. 2023–2024
- **Cohort Representative**, Mechanical and Aerospace Engineering. 2024–2026
- **Organizer**, Charity Trail Run — 150 runners, **600,000 KZT** raised. 2025
- **Volunteer**, HKU Muslim Society and Mereke Charity Foundation. 2025–2026